



PAPER: Statistical Computer Packages
Course Code: STAT-309

TIME ALLOWED: 30 mins.
MAX. MARKS: 10

Attempt this Paper on this Question Sheet only.

OBJECTIVE TYPE

Q1. Read the following items carefully and encircle the correct option listed below at each item.
(One mark for each)

- i) What is the advantage of using SPSS over calculating statistics by hand?
 - a) It equips you with a useful transferable skill.
 - b) It reduces the chance of making errors in your calculations.
 - c) Many researchers use SPSS as it is a recognized software package.
 - d) All of the above.
- ii) In SPSS, what is the "Data Viewer"?
 - a) A table summarizing the frequencies of data for one variable.
 - b) A spreadsheet into which data can be entered.
 - c) A dialog box that allows you to choose a statistical test
 - d) A screen in which variables can be defined and labeled
- iii) How is a variable name different from a variable label?
 - a) It is shorter and less detailed.
 - b) It is abstract and unspecific.
 - c) It is longer and more detailed.
 - d) It refers to codes rather than variables.
- iv) How would you use the drop-down menus in SPSS to generate a frequency table?
 - a) Open the Output Viewer and click: Save As → Pie Chart
 - b) Click on: Analyze; Descriptive Statistics → Frequencies
 - c) Click on: Graphs; Frequencies → Pearson
 - d) Open the Variable Viewer and recode the value labels
- v) In which sub-dialog box can the Chi Square test be found?
 - a) Frequencies → Percentages
 - b) Bivariate → Pearson
 - c) Crosstabs → Statistics
 - d) Gender → Female
- vi) If the null hypothesis is true, then $p < 0.05$ means:
 - a) The obtained result is not due to chance.
 - b) The obtained result is a fairly important effect.
 - c) The obtained result is likely to occur by chance 95% of the time.
 - d) The obtained result is likely to occur by chance less than 5 times in a hundred.
- vii) If you achieved a p-value of 0.04 on a two-tailed test, what would the equivalent one-tailed p-value be?
 - a) 0.2
 - b) 0.002
 - c) 0.06
 - d) 0.02
- viii) Of the below non-parametric tests, which relies on the calculation of ranks?
 - a) Mann Whitney
 - b) Spearman's Rho
 - c) Wilcoxon sign rank test.
 - d) All of the above
- ix) One-way ANOVA is used when:
 - a) analyzing the difference between more than two population means
 - b) analyzing the results of a two tailed tests
 - c) analyzing the results from a large samples
 - d) analyzing the results between two population means
- x) Which of the following tests are used to test for normality?
 - a) A t-test and an ANOVA test
 - b) A Chi-Square test and a Lilliefors test
 - c) An Empirical CDF test and an F-test
 - d) A Quantile-Quantile plot and a p-value test

PAPER: Statistical Computer Packages
Course Code: STAT-309TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPESHORT QUESTIONS

Q2. Define the following:

(4 marks each)

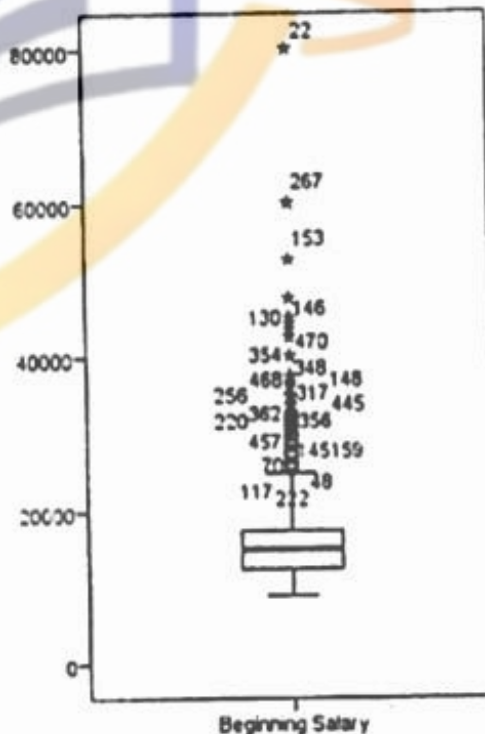
- Enlist various graphical and numerical statistical method to assess the normality of a data
- Note on EDA (Explanatory Data Analysis) tools in SPSS
- Give detail about non-parametric tests which are equivalent to independent and paired samples T-test.
- Categorize the scales of the measurement (nominal, ordinal, interval and ratio) regarding metric and non-metric variables.
- Write short note on SPSS DATA editor window.

SUBJECTIVE

Note: You are required to justify your answers in this section by stating hypothesis, statistics, reference of graphs etc. on your answer sheet.

Q3: a) The following output generated through SPSS from a data containing the Beginning salaries of employees in a company.

	Statistic	Std. Error
Mean	17016.09	361.510
95% Confidence Interval for Mean	Lower Bound 16305.72 Upper Bound 17726.45	
5% Trimmed Mean	16041.71	
Median	15000.00	
Variance	61946945	
Std. Deviation	7870.64	
Minimum	9000	
Maximum	79980	
Range	70980	
Interquartile Range	5168	
Skewness	2.853	.112
Kurtosis	12.390	.224

Tests of Normality

	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	Df	Sig.
Beginning Salary	.252	474	.000	.715	474	.000

a. Lilliefors Significance Correction

Study carefully the above output and answer the following questions: (2+2+2)

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- i) What are the P-values of Kolmogorov-Smirnov and Shapiro-Wilk tests?
- ii) Does this data set follow the normal distribution?
- iii) Set null and alternative hypothesis and accept or reject the normality of the data at 1% level of significance

b) A sample of individuals applying for factory positions at Davis Enterprises revealed the following scores on an eye perception test (X) and a mechanical aptitude test (Y):

Spearman's rho	Correlation Coefficient	1.000	-.191
	Eye Perception Sig. (2-tailed)		.596
	N	10	10
	Correlation Coefficient	-.191	1.000
	Mechanical Aptitude Sig. (2-tailed)	.596	
	N	10	10

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- i) Compute the coefficient of rank correlation.
- ii) At the 0.05 significance level can we conclude that the correlation in the population is different from 0? (2+2)

Q4: a) The annual report of an institute gave the particulars classification of their students smoking habits (Never Smoked, Past Smoker, and Current Smoker) and their class ranks (freshman, sophomore, junior, senior). The following output is generated from the recorded data using SPSS.

			Class Rank				
			Freshman	Sophomore	Junior	Senior	
Smoking Habit	Never Smoked	Count	108	68	60	47	283
		Expected Count	102.2	65.4	69.1	46.3	283.0
		% within Smoking Habit	38.2%	24.0%	21.2%	16.6%	100.0%
		% within Class Rank	77.7%	76.4%	63.8%	74.6%	73.5%
		% of Total	28.1%	17.7%	15.6%	12.2%	73.5%
	Past Smoker	Count	10	4	18	4	36
		Expected Count	13.0	8.3	8.8	5.9	36.0
		% within Smoking Habit	27.8%	11.1%	50.0%	11.1%	100.0%
		% within Class Rank	7.2%	4.5%	19.1%	6.3%	9.4%
		% of Total	2.6%	1.0%	4.7%	1.0%	9.4%
	Current Smoker	Count	21	17	16	12	66
		Expected Count	23.8	15.3	16.1	10.8	66.0
		% within Smoking Habit	31.8%	25.8%	24.2%	18.2%	100.0%
		% within Class Rank	15.1%	19.1%	17.0%	19.0%	17.1%
		% of Total	5.5%	4.4%	4.2%	3.1%	17.1%
Total	Count	139	89	94	63	385	
	Expected Count	139.0	89.0	94.0	63.0	385.0	
	% within Smoking Habit	36.1%	23.1%	24.4%	16.4%	100.0%	
	% within Class Rank	100.0%	100.0%	100.0%	100.0%	100.0%	
	% of Total	36.1%	23.1%	24.4%	16.4%	100.0%	

	Value	Approx. Sig.
Nominal by Nominal Contingency Coefficient	.197	.017
N of Valid Cases	385	

ANOVA^a

Model	Sum of Squares	df	Mean Square	F	Sig.
1 Regression	4.359	3	1.453	10.326	.014 ^b
Residual	.704	5	.141		
Total	5.063	8			

a. Dependent Variable: Paralegal GPA

b. Predictors: (Constant), High School GPA, SAT Verbal, SAT Math

Coefficients^a

Model	Unstandardized Coefficients		Standardized Coefficients	t	Sig.
	B	Std. Error	Beta		
1 (Constant)	-.411	.782		-.525	.622
SAT Verbal	.002	.002	.170	.759	.482
SAT Math	-.002	.002	-.215	-.935	.393
High School GPA	1.201	.296	.945	4.065	.010

a. Dependent Variable: Paralegal GPA

- Compute the value of coefficient of determination from the above output.
- Which independent variable has the strongest correlation with the dependent variable and also test its significance.
- Build a regression model using paralegal GPA as dependent variable and all others as independent variables and also conduct a test of hypothesis for the individual regression coefficients.
- What Paralegal GPA would you estimate for a student when High School GPA score is 2.81, SAT verbal 320 and SAT Math is zero. (2+2+4+2)

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